Snort Deployment and Intrusion Prevention System (IPS) Implementation Report

1. Introduction

Network security is an essential part of modern computing environments. Organizations must protect their systems and networks from unauthorized access, malware, and cyberattacks. One of the most effective tools used in cybersecurity is an **Intrusion Prevention System (IPS)**.

An **Intrusion Prevention System (IPS)** is a network security solution that monitors network traffic and takes action to prevent malicious activities. Unlike an Intrusion Detection System (IDS), which only detects suspicious activity and generates alerts, an IPS actively blocks or prevents malicious traffic from entering the network.

Snort is a widely used open-source network security tool that can operate as both an **Intrusion Detection System (IDS)** and **Intrusion Prevention System (IPS)**. It performs real-time traffic analysis and packet logging to detect various types of attacks.

The purpose of this exercise is to deploy Snort on a Linux virtual machine and analyze how it detects suspicious network activities.

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2. Objectives

The objectives of this lab are:

- To understand the concept of an Intrusion Prevention System (IPS).
- To install and configure Snort on a Linux virtual machine.
- To monitor network traffic using Snort.
- To generate alerts by testing suspicious network activity.
- To analyze logs generated by Snort.

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3. Installation Steps

The following steps were performed to install Snort on the Linux virtual machine.

Step 1: Update the System

First, the system packages were updated to ensure that the latest versions of software were installed.

sudo apt update

sudo apt upgrade

Step 2: Install Snort

Snort was installed using the package manager.

```
sudo apt install snort -y
```

During installation, the system asked for the **HOME_NET configuration**, which represents the local network.

Example configuration:

192.168.1.0/24

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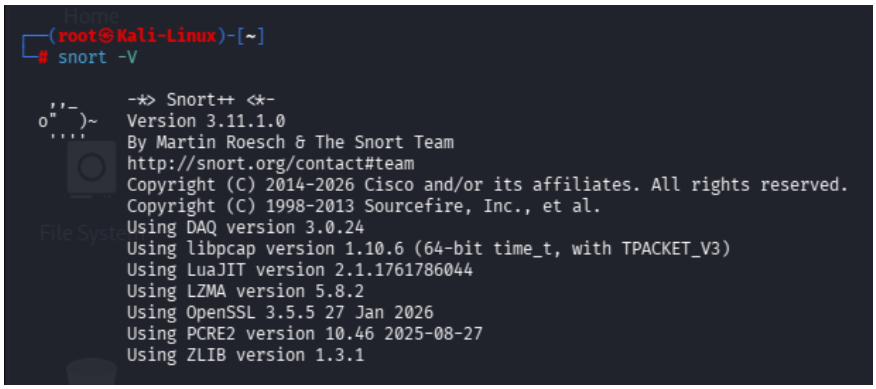
4. Commands Used

Several commands were used during the installation and testing of Snort.

Check Snort Version

```
snort -V
```

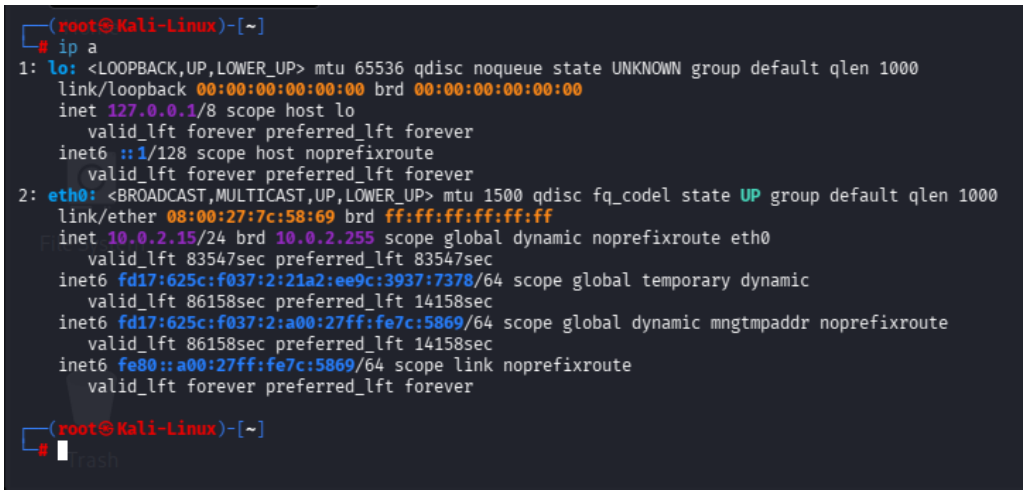
This command verifies whether Snort is installed correctly.



Identify Network Interface

```
ip a
```

This command displays network interfaces such as **eth0** or **ens33**.



Run Snort in IDS Mode

```
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
```

Explanation:

Command	Purpose
-A console	Displays alerts on the terminal
-q	Runs Snort quietly
-c	Specifies configuration file
-i	Specifies network interface

5. Configuration Changes

Snort configuration is mainly done in the following file:

/etc/snort/snort.conf

The configuration file was opened using:

```
sudo nano /etc/snort/snort.conf
```

Setting HOME_NET

The local network was defined in the configuration file:

```
ipvar HOME_NET 192.168.1.0/24
```

Adding Local Rules

Custom detection rules were added in:

/etc/snort/rules/local.rules

Example rule used to detect ping traffic:

```
alert icmp any any -> $HOME_NET any (msg:"Ping Detected"; sid:1000001; rev:1;)
```

This rule generates an alert whenever ICMP ping traffic is detected.

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6. How Snort Works

Snort uses a modular architecture consisting of several components.

1. Packet Decoder

This module captures packets from the network interface and prepares them for analysis.

2. Preprocessors

Preprocessors normalize network traffic and detect anomalies before the detection phase.

Examples include:

- Fragmentation handling

- Protocol normalization
- Packet reassembly

3. Detection Engine

The detection engine compares packets against **Snort rules**. If a packet matches a rule signature, Snort generates an alert.

Rules define patterns that indicate suspicious behavior.

Example rule structure:

```
alert protocol source_ip source_port -> destination_ip destination_port  rule options)
```

4. Logging and Alert System

When suspicious traffic is detected, Snort generates alerts and logs the information.

5. Output Modules

Output modules store logs in files, databases, or monitoring systems.

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7. Test Results

To test Snort, a ping command was executed from another terminal or system.

Example:

```
ping 192.168.1.5
```

Snort detected the traffic and generated alerts in the terminal.

Example alert:

```
[**] [1:1000001:1] Ping Detected [**]
```

Snort logs were stored in the following directory:

```
/var/log/snort/
```

The log files contained detailed information about the detected traffic, including source IP address, destination IP address, protocol type, and timestamp.

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8. Challenges Faced

During the implementation process, several challenges were encountered:

1. Identifying the correct network interface.
2. Ensuring the HOME_NET configuration matched the local network.
3. Properly loading custom rules from the rules directory.
4. Running Snort with appropriate permissions.

These issues were resolved by verifying network configurations and reviewing Snort documentation.